(Please write your Enrollment Number)

0560 119 2023

MID-TERM EXAMINATION

(Course Name : B.Tech AI&ML/ CSE-AI) (Semester : I)

(October, 2023)OFF LINE mode

Subject Code: BAI 103	Subject: Computer Organization & Architecture
Time : 1 ½Hours	Maximum Marks : 30

Note: Q. 1 is compulsory.

Q1		(2.5*4)	
	(a) Explain with the help of the Block diagram, how do we get a Full-adder from Half adders.		
	(b) Simply the expression using K-map : F(A,B,C,D) : Sm : (0,2,3,5,6,7,10,11,14,15)		
	(c) List the 5 different types of "Registers" for Basic Computer. Explain.		
	(d) List out any 5 differences between CISC and RISC architecture.		
Q2	**UNIT-1 Attempt any two part**	(5,5)	
	(a) Using 2's complement representation: i) Subtract 3 from 5 ii) Subtract (-3) from (-5) iii) Add (-5) and (-2) iv) Add 5 and 4 **OR** (a) Simply : (i) $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + ABC$ (ii) $\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$		
	(b) What is a decoder ? Explain how 3x8 decoder can be built using two 2x4 decoders with a logic diagram.		
	(c) Design 4x1 MUX using 2x1 MUX.		
Q3	**UNIT-2 Attempt any two part**	(5,5)	
	(a) Define instruction address format. Explain the various Instruction address formats with the help of examples.		
	(b) Explain any 3 instruction addressing modes with suitable diagram and examples.		
	(c) Explain D_7' I T_3, D_7' I' T_3, D_7 I' T_3 and D_7 I T_3 with the help of Instruction cycle flow Chart.		